

Week 6: Evaluation Layer — Semantic Proxy + Regression Taxonomy

Auto-generated 2026-05-15 11:13 from outputs/week06_bertscore_proxy/ and outputs/week06_analysis/. Do not edit by hand. Re-run python -m src.reporting.build_report --week week06 to refresh.

1. Objective

W3 (paired BM25-vs-reranked generation) and W5 (full-dev rerank) gave us four surface-form deltas (ROUGE-L, BLEU, EM, Token-F1) with paired-bootstrap CIs that all exclude zero. W6 asks two follow-up questions that the surface-form table cannot answer:

1. Does the rerank Δ also show up in a *semantic* similarity metric, or are we just rewarding lexical overlap?
2. The full-dev split has 6,980 paired queries; 1766 are regressions where the reranker brings the relevant passage into top-3 yet the generator scores *worse* than under BM25. What breaks on those queries — retrieval, semantics, or generation?

Both questions are answered cheaply (CPU, ~minutes) and produced under outputs/week06_*/. Neither requires re-running W3/W5.

2. Setup

2.1 BERTScore semantic-proxy

- Inputs: the same two paired prediction files used in the W3 surface-form CI (outputs/week03_generation_bm25_full/predictions.jsonl and outputs/week03_generation_reranked_full/predictions.jsonl).
- Subsample: **3,000 of 6,980 shared qids**, seed 42 (deterministic).
- Scorer: **distilbert-base-uncased** with rescale_with_baseline=True, multi-reference max-F1 per query.
- CI: paired bootstrap, 10,000 resamples, seed 42, 95 % percentile interval.

This is deliberately *not* the canonical roberta-large BERTScore. It is a fast proxy whose only purpose is to answer "is the surface-form Δ also visible in a semantic-similarity scorer?". The full-dev roberta-large pass is listed under §6 *Next*.

2.2 Regression failure taxonomy

- Pool: the **1766 regression queries** identified by `scripts/analyze_generation_rerank.py` (relevant passage in top-3 under rerank, but generator F1 dropped vs BM25).
- Sample: **40** seeded queries, seed 42.
- Labels are produced by a deterministic rule cascade (`scripts/regression_failure_taxonomy.py`); first-match wins: `truncation_short`, `truncation_midword`, `topic_drift`, `extractive_passage_bias`, `semantic_mismatch`.

This is a *triage*, not a citation-grade classifier — the value is in the headline split, not the per-row labels.

3. BERTScore proxy results

Metric	BM25 (mean)	Reranked (mean)	Δ (rerank – BM25)	95 % CI	p (two- sided)
BERTScore-F1 (DistilBERT proxy)	0.2192	0.3920	+0.1728	[+0.1608, +0.1850]	< 0.001

Per-query split on the 3,000-pair subsample:

Outcome	Rate
Rerank strictly better	64.8 %
Tie	5.7 %
BM25 strictly better	29.5 %

The semantic Δ (+0.1728) lands within a hair of the surface-form Token-F1 Δ already published in the W3 paired-bootstrap table (+0.1711) and the ROUGE-L Δ (+0.1742). **The rerank gain is not a surface-form artefact** — a semantic-similarity scorer picks it up at the same magnitude.

4. Regression taxonomy

40-query seeded triage of the 1766-strong regression bucket.

Failure mode	Count	% of sample
<code>truncation_midword</code>	22	55.0 %
<code>truncation_short</code> (≤ 3 tok)	14	35.0 %
<code>topic_drift</code>	2	5.0 %
<code>extractive_passage_bias</code>	2	5.0 %
<code>semantic_mismatch</code>	0	0.0 %

90.0 % of regressions are generation-side truncation, not retrieval or semantic failures. The cross-encoder gets the relevant passage into the generator's window; the generator then cuts the answer at `max_new_tokens=64` or extracts a title-like fragment. The cheapest intervention by far is raising the generator's token budget — see §6 *Next*.

`semantic_mismatch` did not pick up any sampled rows under the rule cascade. Per the script's documentation that bucket is the residual catch-all; its emptiness here is consistent with the BERTScore result in §3 (the rerank gain is real on a semantic scorer).

5. Limitations

- **DistilBERT, not roberta-large.** The headline $\Delta + \text{CI}$ in §3 is a *proxy*. For citation in a cross-paper comparison, re-run `scripts/bertscore_paired_eval.py` with `--model-type roberta-large --n-pairs 0` (full dev, ~hours of CPU). The proxy is calibrated only for "is the surface-form Δ also visible in a semantic scorer", not for absolute BERTScore numbers.
- **Sampled, not full-dev BERTScore.** $3,000/6,980 = 43.0\%$ of the shared qid set; full-dev is trivially obtainable but does not change the qualitative answer.
- **Heuristic taxonomy.** The five labels in §4 are deterministic rules, not a learned classifier. Labels are intentionally coarse; the load-bearing claim is the truncation share, not the per-row attribution.
- **Single sampled triage.** 40 of 1766 = $\sim 17\%$ of the regression bucket. Seed-stable, but a different seed would shift the exact percentages by a few points.

6. Next

- **Bump T5-small max_new_tokens from 64 → 128.** The taxonomy says $\sim 90\%$ of regressions are output truncation, which makes this the highest-expected-payoff cheap fix. Re-run the paired W3 generation comparison and the regression taxonomy after the change.
- **Full-dev BERTScore with roberta-large.** Convert the §3 proxy into a citation-grade evaluation. Same script, two CLI flags.
- **Failure-mode classifier upgrade.** Once truncation is mitigated, the residual regression bucket will be dominated by topic-drift / extractive-bias / semantic-mismatch; a learned classifier (rather than the current rule cascade) is appropriate at that point.

7. Reproduce

```
# (1) Generation on full dev/small – done in W3, but listed here
# for
# completeness; mutually restricted to the shared qid set:
python experiments/run_generation_baseline.py \
  --input-run outputs/week02_bm25/run.tsv \
  --output-dir outputs/week03_generation_bm25_full \
  --retrieval-source bm25 \
  --restrict-to-run outputs/week05_reranker_full/run.tsv \
```

```
--num-eval-queries 9999

python experiments/run_generation_baseline.py \
  --input-run outputs/week05_reranker_full/run.tsv \
  --output-dir outputs/week03_generation_reranked_full \
  --retrieval-source reranked \
  --restrict-to-run outputs/week02_bm25/run.tsv \
  --num-eval-queries 9999

# (2) Bucket / retrieval-flag analysis – emits regression bucket
#      and
# summary.json under outputs/week06_analysis/:
python scripts/analyze_generation_rerank.py \
  --bm25-dir outputs/week03_generation_bm25_full \
  --reranked-dir outputs/week03_generation_reranked_full \
  --output-dir outputs/week06_analysis

# (3) BERTScore semantic-proxy CI (~3 min CPU on the 3 000-pair
#      subsample):
python scripts/bertscore_paired_eval.py \
  --bm25-dir outputs/week03_generation_bm25_full \
  --reranked-dir outputs/week03_generation_reranked_full \
  --output-dir outputs/week06_bertscore_proxy \
  --n-pairs 3000

# (4) Regression failure taxonomy (40-query seeded triage):
python scripts/regression_failure_taxonomy.py
```